

Writing the Inverse of a Linear Function - Practice

Algebra II | Unit 1 | Lesson 11 Practice | TEK A2.2.B

Objective: I can write the inverse of a linear function by swapping x and y and solving for y .

Section 1 — Recall

1. Solve $y = 5x - 8$ for x .

$$x = \underline{\hspace{2cm}}$$

2. Find the inverse of $f(x) = x + 9$.

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

Section 2 — Apply

3. Find the inverse of $f(x) = 4x - 2$.

$$f^{-1}(x) = \underline{\hspace{2cm}}$$

4. Find the inverse of $f(x) = -x + 6$, and identify what's special about this particular function.

$$f^{-1}(x) = \underline{\hspace{2cm}} \quad \underline{\hspace{2cm}}$$

Section 3 — Challenge

5. Slopes of f and f^{-1} are reciprocals. If $f(x) = (2/3)x + 5$, predict the slope of f^{-1} without doing the swap-and-solve steps, then verify by actually finding f^{-1} .

Predicted slope: _____ $f^{-1}(x) =$ _____

6. A vending machine converts quarters to store credit using $f(q) = 0.25q$ (q = quarters, $f(q)$ = dollars). Find f^{-1} , explain what it does in words, and use it to find how many quarters give \$3.75 of credit.

$$f^{-1}(x) = \underline{\hspace{2cm}} \quad f^{-1}(3.75) = \underline{\hspace{2cm}}$$

Section 4 — Explain It

7. A classmate says: "Since $f(x) = 3x$ has an inverse, every linear function must have an inverse that is also a function, no exceptions." Is the classmate correct? Use a horizontal line, like $f(x) = 7$, to test the claim.
